Components of a Data Warehouse

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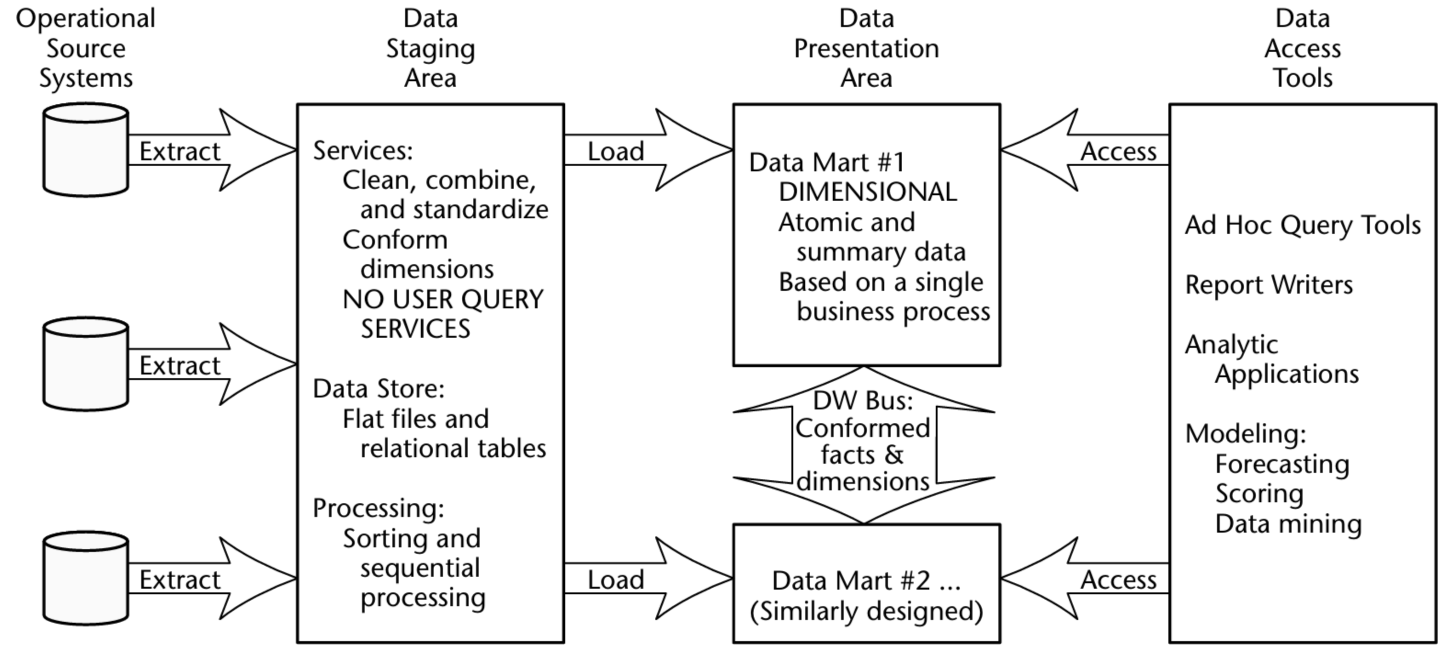

Figure 1.1 Basic elements of the data warehouse.

1. Operational Source Systems

- Systems that store **daily transactions (OLTP)**

Features:

- Fast for insert/update
- Contains current data (not much history)
- Not for analysis

We only **extract data** from here

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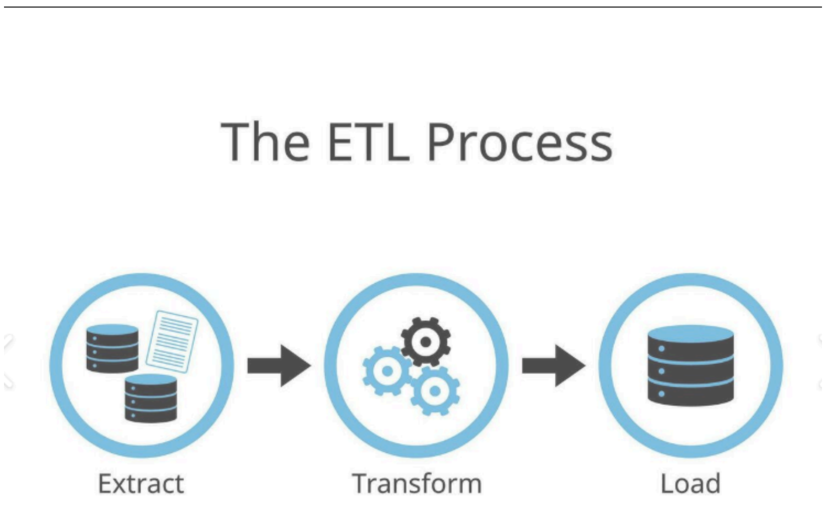
2. Data Staging Area (ETL)

The data staging area of the data warehouse is both a storage area and a set of processes commonly referred to as extract-transformation-load (ETL).

The data staging area is everything between the operational source systems and the data presentation area. It is somewhat analogous to the kitchen of a restaurant, where raw food products are transformed into a fine meal.

Steps:

- Extract → get data
- Transform → clean & fix
  - Data Cleaning
  - Standardization
  - Combine data from multiple sources
  - Deduplication
  - Assigning Keys (surrogate keys)
- Load → prepare for DWH



What we do:

- cleaning
- combining data
- remove duplicates
- standardize

Users **cannot access** staging

No queries happen here

Main tasks: clean, combine, deduplicate, standardize

Normalization in Staging?

- Normalized tables can **support ETL**, but:
  - They are **not for users or queries**
  - Once a table is queryable → it becomes part of the **presentation area**
- The **real goal** = build a presentation layer for **analysis & decision making**

Debate: Should staging be normalized?

- Wrong: Normalized tables = Enterprise DWH
- Correct: **Enterprise DWH = Staging + Presentation**

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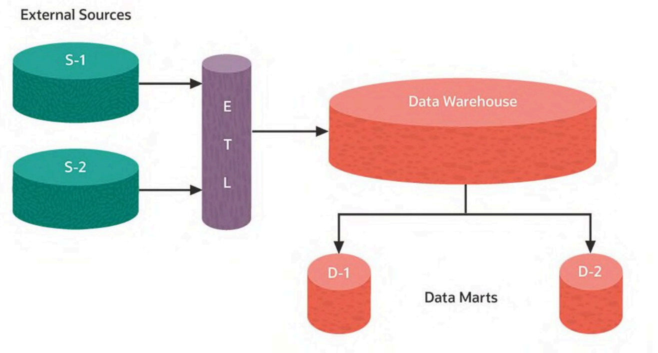
3. Data Presentation Area (Data Marts)

The **data presentation area** = where data is **organized, stored, and available for users**, report writers, and analytical tools.

- The **business only sees this area**.
- Often referred to as a series of **integrated data marts**.
- Each **data mart** = wedge of overall data, usually covering **one business process**.

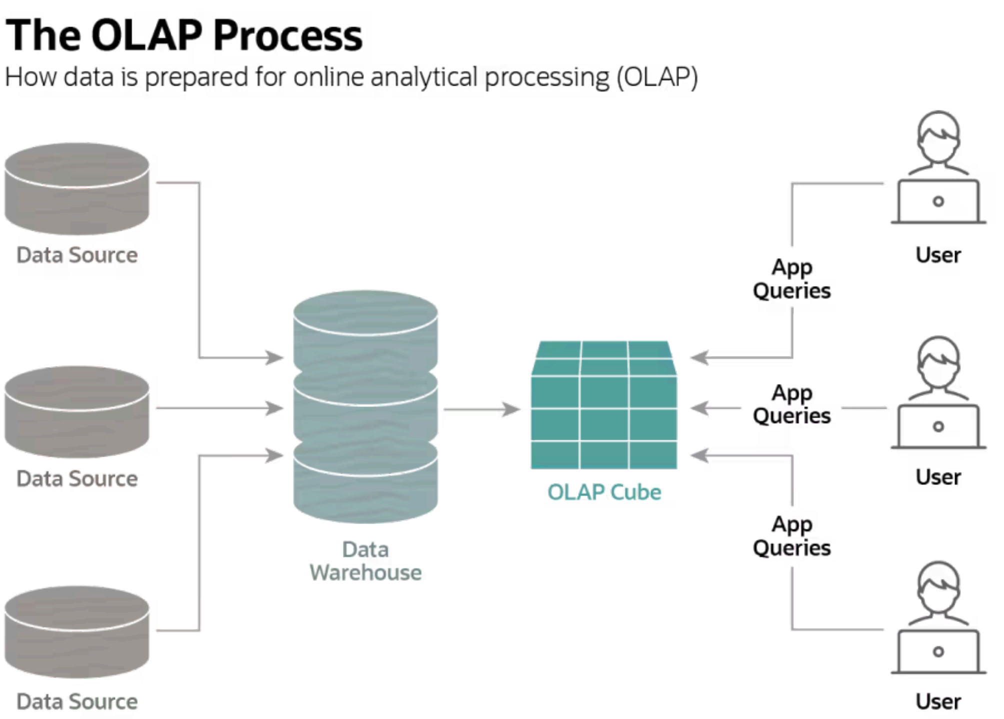


data is stored in **dimensional model (star schema)**

- **Dimensional Modeling**
  - Must use **dimensional schemas** for simplicity and understandability.
  - Users think in **dimensions** (e.g., product, market, time) → cube structure.
  - Supports **slicing & dicing** for analysis.
  - Avoid overly complex **normalized models** in presentation → hard to query, understand, and slow performance.
- **Atomic Data First**
  - Data marts must store **detailed, granular data**.
  - Summaries/aggregates are optional but **cannot replace atomic data**.
  - Users need precise data for unpredictable ad-hoc queries.

"granular" (or **granularity**) refers to **how detailed or fine the data is**.

- **Conformed Dimensions & Facts**
  - All data marts must **share dimensions & facts** → enables integration.
  - Prevents **stovepipe data marts** (isolated, incompatible).
  - Foundation of **Data Warehouse Bus Architecture** → allows **decentralized but integrated development**.
- **Schema Types**
  - **Relational DWH** → uses **star schemas**
  - Logical design = same (dimensions + facts), physical differs.
  - **OLAP / Multidimensional** → uses **cubes**



OLAP works by extracting data from multiple sources and formatting it into cubes, which can then be analyzed from multiple points of view. Multiple cubes can be nested, creating multidimensional "hypercubes".

> An **OLAP Cube** is a **multi-dimensional structure** that stores data for fast analysis. It allows users to **slice, dice, drill down, and roll up** data across different dimensions like **Product, Time, and Location**. It is used for interactive business analysis and reporting.

- **Updates**
  - Modern data marts **can be updated** (managed loads, not transactions).
  - Changes = labels, hierarchies, ownership, or corrections.

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4. Data Access Tools

tools for users

- reports
- dashboards
- BI tools

users don't write complex SQL

they use simple tools

Additional Considerations in Data Warehousing

1. Metadata

- **Definition:** Metadata = data about data (like an encyclopedia for your data warehouse).
- **Purpose:** Helps technical teams, admins, and business users understand, manage, and use the data.
- **Types:**
  - **Source Metadata:** Source schemas, extraction instructions.
  - **Staging Metadata:** Transformation rules, cleansing rules, ETL schedules, table layouts.
  - **Warehouse Metadata:** Indexes, partitions, security, views.
  - **Presentation Metadata:** Business names, column definitions, access logs, templates.
- **Key Idea:** Metadata is crucial. It organizes, documents, and supports proper use of the warehouse.

2. Operational Data Store (ODS)

- **Definition:** A temporary store of operational data, slightly integrated and frequently updated.
- **Purpose:** Supports:
  - Operational reporting (tactical decisions)
  - Real-time interactions (e.g., CRM systems like checking a service history or travel itinerary)
- **Characteristics:**
  - Limited queries (fixed structures)
  - Does **not** store historical aggregates or full descriptive data
  - Can be a separate system or a "hot" partition inside the main data warehouse
- **Key Idea:**
  - ODS is optional. Only needed if operational systems or the warehouse cannot meet immediate business needs.
  - Granular atomic data should be considered part of the **data warehouse presentation area**, not a separate ODS.